

One tree, N ringwidth measurements r_w for each year x_n .

For now, only a long-term Gaussian process:

$$\forall x_n \quad \log r_w(x_n) \sim \text{normal}(\mu + f_{\text{long}}(x_n), \sigma^2) \quad (1)$$

with:

$$f_{\text{long}}^{\vec{}} \sim \text{multinormal}(\vec{0}, K_{\text{long}}) \quad (2)$$

Each element k of the matrix $K_{\text{long}}^{N \times N}$ is defined by the exponentiated quadratic function:

$$\forall i, j \in \llbracket 1, N \rrbracket \quad k_{i,j} = \gamma_{\text{long}}^2 \exp \left(-\frac{1}{2} \left(\frac{|x_j - x_i|}{\rho_{\text{long}}} \right)^2 \right) \quad (3)$$

with γ_{long} the marginal deviation and ρ_{long} the length scale

Note that equations (1) and (2) are equivalent to:

$$\log r_w^{\vec{}} \sim \text{multinormal}(\vec{\mu}, \Sigma + K_{\text{long}}) \quad (4)$$

where:

$$\vec{\mu} = \begin{bmatrix} \mu \\ \mu \\ \vdots \\ \mu \end{bmatrix} \quad \Sigma = \begin{bmatrix} \sigma^2 & 0 & \dots & 0 \\ 0 & \sigma^2 & \dots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & \sigma^2 \end{bmatrix} \quad (5)$$

For one tree, it's pretty straightforward to add a short-term Gaussian process:

$$\log r_w^{\vec{}} \sim \text{multinormal}(\vec{\mu}, \Sigma + K_{\text{long}} + K_{\text{short}}) \quad (6)$$

This short-term Gaussian process is also defined by an exponentiated quadratic function with parameters γ_{short} and ρ_{short} .

In this case, you can write only one Gaussian process, with one matrix $K_{\text{long+short}}$ with two exponentiated quadratic functions:

$$\forall i, j \in \llbracket 1, N \rrbracket \quad k_{i,j} = \gamma_{\text{long}}^2 \exp \left(-\frac{1}{2} \left(\frac{|x_j - x_i|}{\rho_{\text{long}}} \right)^2 \right) + \gamma_{\text{short}}^2 \exp \left(-\frac{1}{2} \left(\frac{|x_j - x_i|}{\rho_{\text{short}}} \right)^2 \right) \quad (7)$$

Several trees $t \in \llbracket 1, T \rrbracket$ on the same stand, with the same number of ringwidth measurements:

$$\forall t \in \llbracket 1, T \rrbracket \quad \log r_w^{\vec{}}(t) \sim \text{multinormal}(\vec{\mu}, \Sigma + K_{\text{long+short}}) \quad (8)$$